

Toshiba Bi-CD Integrated Circuit Silicon Monolithic

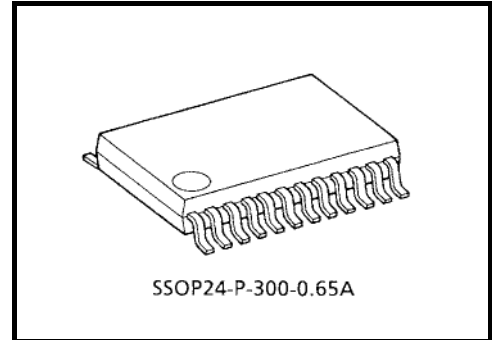
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Driver IC for Dual DC motor

TB6612FNG is a driver IC for DC motor with output transistor in LD MOS structure with low ON-resistor. Two input signals, IN1 and IN2, can choose one of four modes such as CW, CCW, short brake, and stop mode.

Features

- Power supply voltage ; $V_M=15V$ (Max.)
- Output current ; $I_{out}=1.2A(ave) / 3.2A$ (peak)
- Output low ON resistor ; 0.5Ω (upper+lower Typ. @ $V_M \geq 5V$)
- Standby (Power save) system
- CW/CCW/short brake/stop function modes
- Built-in thermal shutdown circuit and low voltage detecting circuit
- Small faced package (SSOP24 : 0.65mm Lead pitch)
- Response to Pb free packaging



質量: 0.14 g (標準)

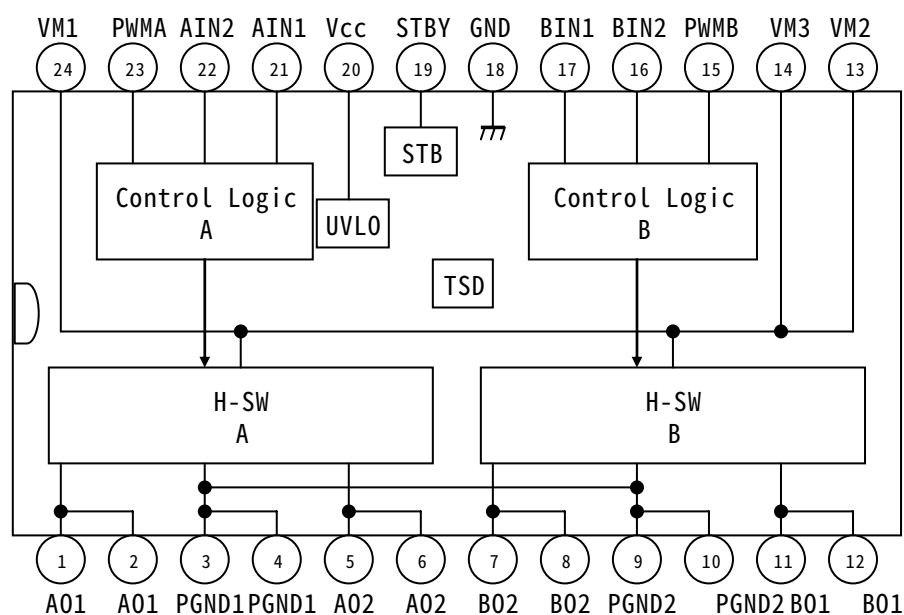
- * This product has a MOS structure and is sensitive to electrostatic discharge. When handling this product, ensure that the environment is protected against electrostatic discharge by using an earth strap, a conductive mat and an ionizer. Ensure also that the ambient temperature and relative humidity are maintained at reasonable levels.

The TB6612FNG is a Pb-free product.
The following conditions apply to solderability:

*Solderability

1. Use of Sn-37Pb solder bath
*solder bath temperature = 230°C
*dipping time = 5 seconds
*number of times = once
*use of R-type flux
2. Use of Sn-3.0Ag-0.5Cu solder bath
*solder bath temperature = 245°C
*dipping time = 5 seconds

Block Diagram



Pin Functions

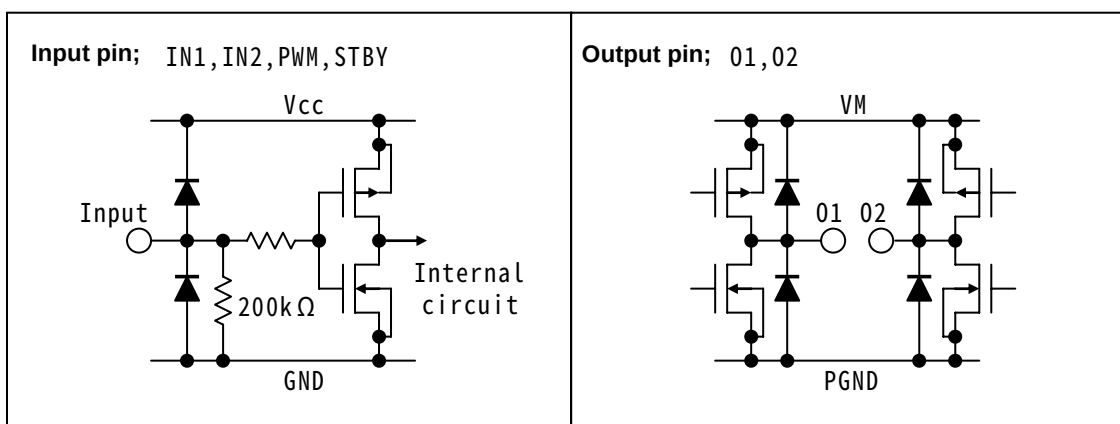
Pin NO.	Symbol	I/O	Remarks
1	AO1	O	chA output1
2	AO1		
3	PGND1	—	Power GND 1
4	PGND1		
5	AO2	O	chA output2
6	AO2		
7	BO2	O	chB output2
8	BO2		
9	PGND2	—	Power GND 2
10	PGND2		
11	BO1	O	chB output1
12	BO1		
13	VM2	—	Motor supply (2.5V~13.5V)
14	VM3		
15	PWMB	I	chB PWM input / 200k Ω pull-down at internal
16	BIN2	I	chB input2 / 200k Ω pull-down at internal
17	BIN1	I	chB input1 / 200k Ω pull-down at internal
18	GND	—	Small signal GND
19	STBY	I	“L”=standby / 200k Ω pull-down at internal
20	Vcc	—	Small signal supply (2.7V~5.5V)
21	AIN1	I	chA input1 / 200k Ω pull-down at internal
22	AIN2	I	chA input2 / 200k Ω pull-down at internal
23	PWMA	I	chA PWM input / 200k Ω pull-down at internal
24	VM1	—	Motor supply (2.5V~13.5V)

Absolute Maximum Ratings (Ta = 25°C)

Characteristics	Symbol	Rating	Unit	Remarks
Supply voltage	VM	15	V	
	Vcc	6		
Input voltage	VIN	-0.2~6	V	IN1,IN2,STBY,PWM pins
Output voltage	Vout	15	V	01,02 pins
Output current	Iout	1.2	A	Per 1ch
	Iout (peak)	2		tw=20ms Continuous pulse, Duty≤20%
		3.2		tw=10ms Single pulse
Power dissipation	PD	0.78	W	IC only
		0.89		50×50 t=1.6(mm) Cu≥40% in PCB mounting
		1.36		76.2×114.3 t=1.6(mm) Cu≥30% in PCB monting
Operating temperature	Topr	-20~85	°C	
Storage temperature	Tstg	-55~150	°C	

Operating Range (Ta=-20~85°C)

Characteristics	Symbol	Min	Typ.	Max	Unit	Remarks
Supply voltage	Vcc	2.7	3	5.5	V	
	VM	4.5	5	13.5	V	
Output current (H-SW)	Iout	---	---	1.0	A	VM≥5V
		---	---	0.4		5V>VM≥4.5V
Switching frequency	fPWM	---	---	100	kHz	

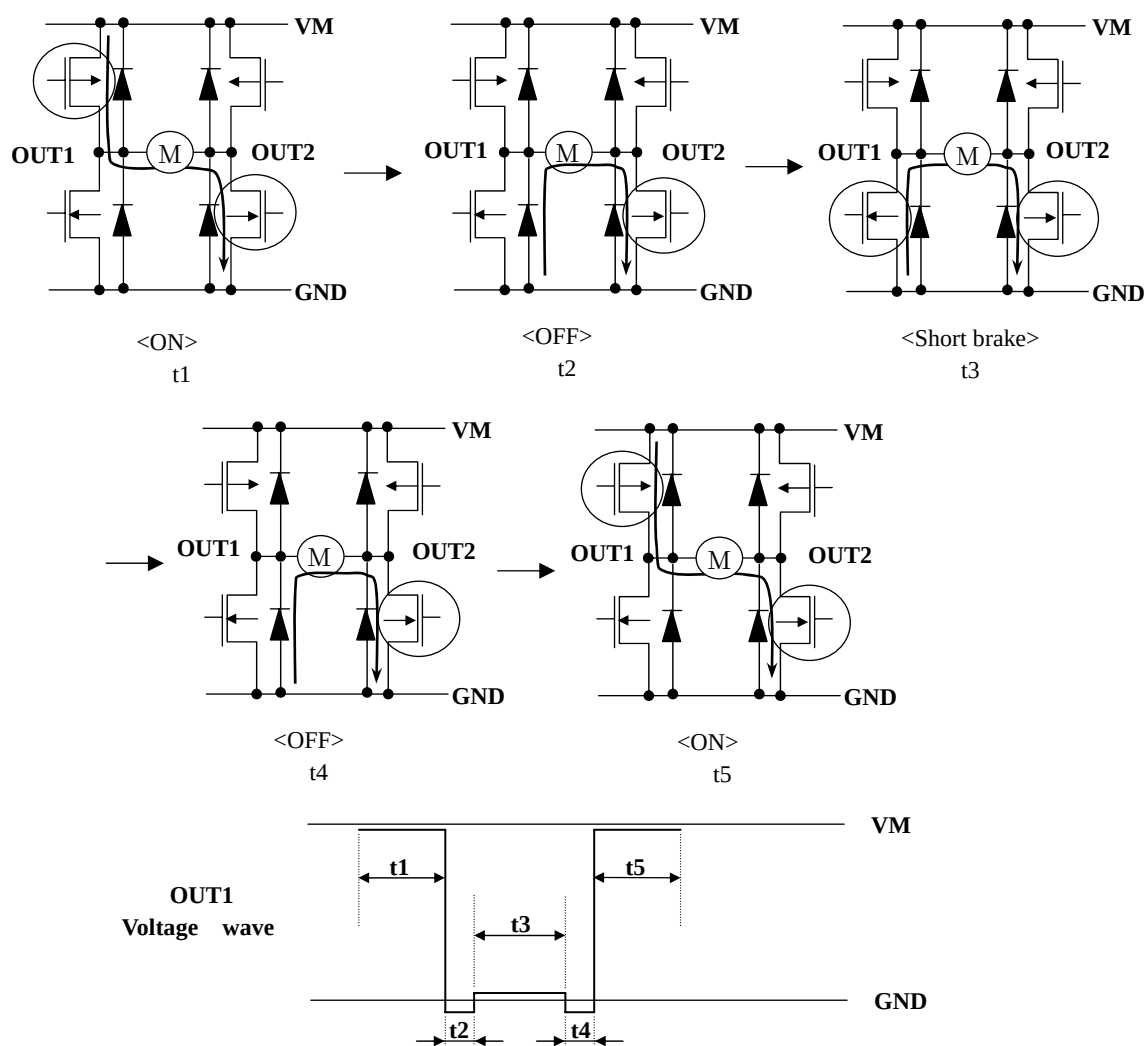


H-SW Control Function

Input				Output		
IN1	IN2	PWM	STBY	OUT1	OUT2	Mode
H	H	H/L	H	L	L	Short brake
L	H	H	H	L	H	CCW
		L	H	L	L	Short brake
H	L	H	H	H	L	CW
		L	H	L	L	Short brake
L	L	H	H	OFF (High impedance)		Stop
H/L	H/L	H/L	L	OFF (High impedance)		Standby

H-SW Operating Description

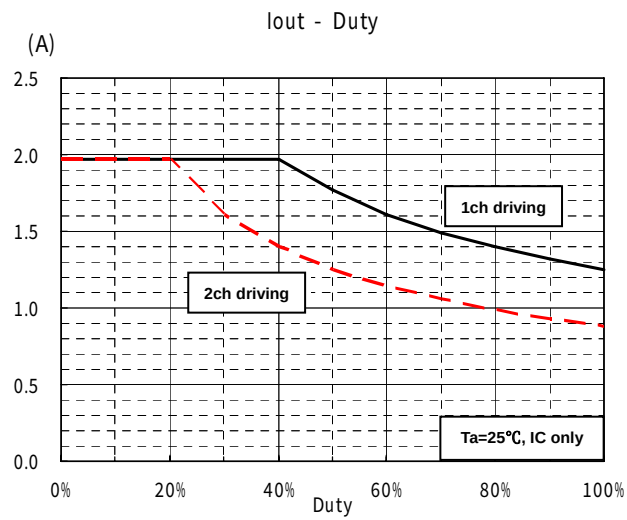
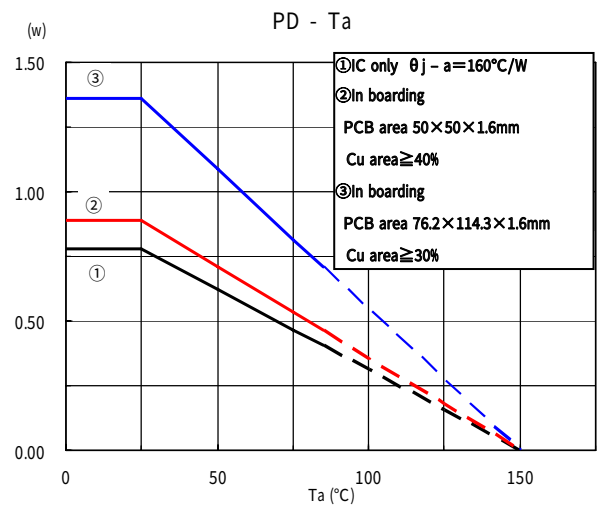
- To prevent penetrating current, dead time t_2 and t_4 is provided in switching to each mode in the IC.



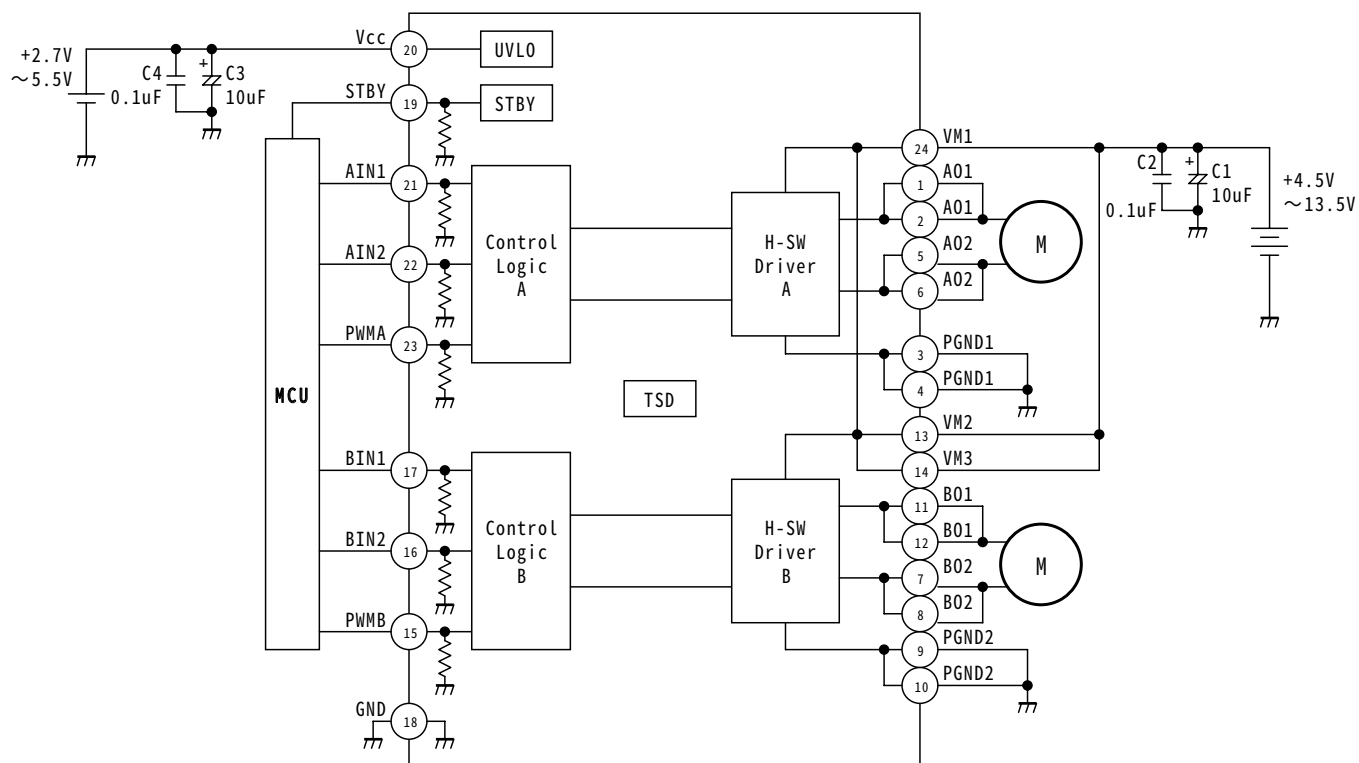
Electrical Characteristics (unless otherwise specified, Ta = 25°C, Vcc=3V, VM=5V)

Characteristics	Symbol	Test Condition	Min	Typ.	Max	Unit
Supply current	Icc(3V)	STBY=Vcc=3V, VM=5V	---	1.1	(1.8)	mA
	Icc(5.5V)	STBY=Vcc=5.5V, VM=5V	---	1.5	2.2	
	Icc(STB)	STBY=0V	---	---	1	μA
	IM(STB)		---	---	1	
Control input voltage	VIH		Vcc×0.7	---	Vcc+0.2	V
	VIL		-0.2	---	Vcc×0.3	
Control input current	IIH	VIN=3V	5	15	25	μA
	IIL	VIN=0V	---	---	1	
Standby input voltage	VIH(STB)		Vcc×0.7	---	Vcc+0.2	V
	VIL(STB)		-0.2	---	Vcc×0.3	
Standby input current	IIH(STB)	VIN=3V	5	15	25	μA
	IIL(STB)	VIN=0V	---	---	1	
Output saturating voltage	Vsat(U+L)1	Io=1A, Vcc=VM=5V	---	0.5	(0.7)	V
	Vsat(U+L)2	Io=0.3A, Vcc=VM=5V		0.15	(0.21)	
Output leakage current	IL(U)	VM=Vout=15V	---	---	1	μA
	IL(L)	VM=15V, Vout=0V	-1	---	---	
Regenerative diode VF	VF(U)	IF=1A	---	1	1.1	V
	VF(L)		---	1	1.1	
Low voltage detecting voltage	UVLD	(Designed value)	---	1.9	---	V
Recovering voltage	UVLC		---	2.2	---	
Response speed	tr	(Designed value)	---	24	---	ns
	tf		---	41	---	
	Dead time	H to L	---	(50)	---	
		L to H	---	(230)	---	
Thermal shutdown circuit operating temperature	TSD	(Designed value)	---	175	---	°C
Thermal shutdown hysteresis	Δ TSD		---	20	---	

Target characteristics



Typical Application Diagram

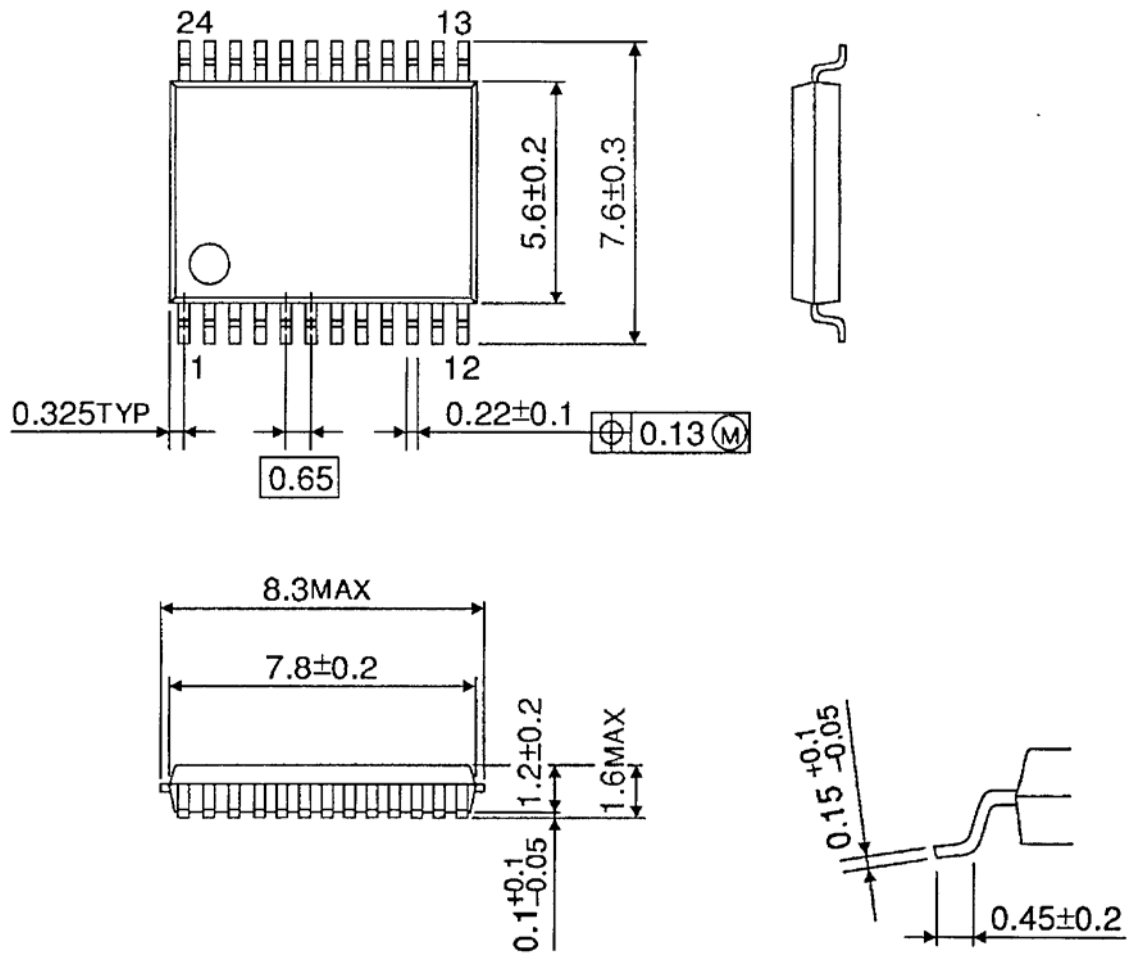


Note: Condensers for noise absorption (C1, C2, C3, and C4) should be connected as close as possible to the IC.

Package Dimennsions

SSOP24-P-300-0.65A

Unit : mm



Weght: 0.14 g (typ)

Notes on Contents

1. Block Diagrams

Some of the functional blocks, circuits, or constants in the block diagram may be omitted or simplified for explanatory purposes.

2. Equivalent Circuits

The equivalent circuit diagrams may be simplified or some parts of them may be omitted for explanatory purposes.

3. Timing Charts

Timing charts may be simplified for explanatory purposes.

4. Application Circuits

The application circuits shown in this document are provided for reference purposes only. Thorough evaluation is required, especially at the mass production design stage.

Toshiba does not grant any license to any industrial property rights by providing these examples of application circuits.

5. Test Circuits

Components in the test circuits are used only to obtain and confirm the device characteristics. These components and circuits are not guaranteed to prevent malfunction or failure from occurring in the application equipment.

IC Usage Considerations

Notes on handling of ICs

- [1] The absolute maximum ratings of a semiconductor device are a set of ratings that must not be exceeded, even for a moment. Do not exceed any of these ratings.
Exceeding the rating(s) may cause the device breakdown, damage or deterioration, and may result injury by explosion or combustion.
- [2] Use an appropriate power supply fuse to ensure that a large current does not continuously flow in case of over current and/or IC failure. The IC will fully break down when used under conditions that exceed its absolute maximum ratings, when the wiring is routed improperly or when an abnormal pulse noise occurs from the wiring or load, causing a large current to continuously flow and the breakdown can lead smoke or ignition. To minimize the effects of the flow of a large current in case of breakdown, appropriate settings, such as fuse capacity, fusing time and insertion circuit location, are required.
- [3] If your design includes an inductive load such as a motor coil, incorporate a protection circuit into the design to prevent device malfunction or breakdown caused by the current resulting from the inrush current at power ON or the negative current resulting from the back electromotive force at power OFF. IC breakdown may cause injury, smoke or ignition.
Use a stable power supply with ICs with built-in protection functions. If the power supply is unstable, the protection function may not operate, causing IC breakdown. IC breakdown may cause injury, smoke or ignition.
- [4] Do not insert devices in the wrong orientation or incorrectly.
Make sure that the positive and negative terminals of power supplies are connected properly. Otherwise, the current or power consumption may exceed the absolute maximum rating, and exceeding the rating(s) may cause the device breakdown, damage or deterioration, and may result injury by explosion or combustion.
In addition, do not use any device that is applied the current with inserting in the wrong orientation or incorrectly even just one time.

Points to remember on handling of ICs**(1) Thermal Shutdown Circuit**

Thermal shutdown circuits do not necessarily protect ICs under all circumstances. If the thermal shutdown circuits operate against the over temperature, clear the heat generation status immediately.

Depending on the method of use and usage conditions, such as exceeding absolute maximum ratings can cause the thermal shutdown circuit to not operate properly or IC breakdown before operation.

(2) Heat Radiation Design

In using an IC with large current flow such as power amp, regulator or driver, please design the device so that heat is appropriately radiated, not to exceed the specified junction temperature (T_J) at any time and condition. These ICs generate heat even during normal use. An inadequate IC heat radiation design can lead to decrease in IC life, deterioration of IC characteristics or IC breakdown. In addition, please design the device taking into consideration the effect of IC heat radiation with peripheral components.

(3) Back-EMF

When a motor rotates in the reverse direction, stops or slows down abruptly, a current flows back to the motor's power supply due to the effect of back-EMF. If the current sink capability of the power supply is small, the device's motor power supply and output pins might be exposed to conditions beyond maximum ratings. To avoid this problem, take the effect of back-EMF into consideration in system design.

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